PAPER_943: SMBH Binary Merger Phonon Modulation

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 213 **Source:** smbh_binary_mergers.py (SMBHBinaryMergerPhonon) **Calculator:** SMBHBinaryMergerPhononCalc (CP4 #527) **CVW:** v2.0.0 compliant

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Abstract

We extend UQFF phonon modulation to supermassive black hole binary mergers in the LISA gravitational-wave band (1–100 mHz). The merger power is expressed as $P_{\text{merger}}(\Gamma) = P_{\text{GR}} \cdot (1 + M_{\text{merger}}(\Gamma))$, where the phonon modulation factor M_{merger} encodes the 26-channel buoyancy response. The Fitchett mass-ratio formula governs the recoil kick, and strain damping ranges from 47% ($q = 1$) to 66.7% ($q \to 0$) via the relation $D_{\text{total}}(q) = 0.333 + 0.197(1 - q)$.

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1. Merger Power

$$P_{\text{merger}}(\Gamma) = P_{\text{GR}} \cdot \left(1 + M_{\text{merger}}(\Gamma)\right)$$

$$P_{\text{GR}} = 3.6 \times 10^{49}\,(4\eta)^2\ \text{W}$$

where $\eta = M_1 M_2/(M_1 + M_2)^2$ is the symmetric mass ratio.

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2. Chirp Mass

$$M_c = M_{\text{total}} \cdot \eta^{3/5}$$

For $M_1 = 10^8\, M_\odot$, $M_2 = 5 \times 10^7\, M_\odot$: $q = 0.5$, $M_c \approx 6.53 \times 10^7\, M_\odot$.

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3. Strain Damping

$$D_{\text{total}}(q) = 0.333 + 0.197 \cdot (1 - q)$$

q	D_{total}	Damping
1.0	0.333	66.7%
0.5	0.432	56.8%
0.1	0.510	49.0%

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4. Source Data

- **File:** smbh_binary_mergers.py
- **Session:** 213
- **CP4 Class:** SMBHBinaryMergerPhononCalc (#527)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Fitchett, M.J. (1983) – MNRAS, 203, 1049
3. LISA Consortium (2023) – arXiv:2402.07571